

Telecom Customer Churn — Survival Analysis by Service Tier

Domain: Telecom / Subscription Business | Tools: SPSS | Data: 1,000 Customers (Kaplan-Meier Survival Analysis)

Summary

Objective

Determine how customer tenure before churn ('survival time') differs across telecom service tiers, to identify which customer segment churns fastest and needs the most urgent retention attention.

Concepts / Methods Used

Kaplan-Meier Survival Analysis, Censored Data Handling, Survival & Hazard Function Estimation, Log-Rank (Mantel-Cox) Test, Breslow (Generalized Wilcoxon) Test, Tarone-Ware Test for group comparison.

Key Output

- Sample: 1,000 telecom customers across 4 service categories (Basic, E-Service, Plus, Total Service); 274 churn events observed, 726 customers censored (still active / not yet churned at study end)
- Median survival time varies sharply by tier: Basic Service = 22 months, E-Service = 45 months, Total Service = 61 months, Plus Service = 69 months
- Basic Service customers churn roughly 3x faster than Plus Service customers (22 vs 69 months median tenure)
- Difference in survival distributions across the 4 categories is highly statistically significant — confirmed by all three tests (Log-Rank $\chi^2=50.495$, Breslow $\chi^2=51.573$, Tarone-Ware $\chi^2=52.163$; all $df=3$, $p<.001$)
- Plus Service has the highest censoring rate (84.3%) — most Plus customers had not churned by the end of the observation window, consistent with strong retention

Business Actionables

- Treat Basic Service as the highest-priority retention segment — its median tenure (22 months) is the shortest by a wide margin
- Time retention outreach to occur before month ~20–22 for Basic Service customers, since that is where the churn risk concentrates
- Use tier upgrade incentives (Basic → E-Service or Plus) as a retention lever — higher tiers show 2–3x longer customer lifetime
- Since the difference between tiers is statistically confirmed (not random noise), retention budget can be allocated by tier with confidence rather than applying a blanket, one-size-fits-all retention program

Detailed Analysis — Actual SPSS Output, Explained, With Actionables

Output 1 — Case Processing Summary

Customer Category	Total N	N of Events	Censored N	Censored %
Basic Service	266	83	183	68.8%
E-Service	217	59	158	72.8%
Plus Service	281	44	237	84.3%
Total Service	236	88	148	62.7%
Overall	1000	274	726	72.6%

Reproduced from actual SPSS Kaplan-Meier output: Case Processing Summary by customer category

What this is: A breakdown of the 1,000-customer sample by service category, showing how many customers actually churned (events) versus how many were censored (still active or otherwise not observed to churn) during the study window.

Layman meaning: Plus Service has the highest censoring rate (84.3%) — meaning the large majority of Plus customers were still subscribed when the study ended, a strong sign of retention. Total Service has the lowest censoring rate (62.7%), meaning a higher share of these customers had already churned. This is the first hint that plan tier and customer stickiness are related.

Actionable for the user: Censoring rate alone is a quick health check per segment — any tier with unusually low censoring (high churn-within-window) deserves a closer look before even running the full survival model.

Output 2 — Survival Time Summary (Mean vs Median)

Customer Category	Mean Survival (Months)	Mean Std. Error	Median Survival (Months)	Median Std. Error
Basic Service	48.422	1.98	22	4.451
E-Service	58.146	1.565	45	5.515
Plus Service	62.974	1.261	69	.
Total Service	50.55	1.783	61	3.746

Reproduced from actual SPSS Kaplan-Meier output: Means and Medians for Survival Time

What this is: The estimated average (mean) and midpoint (median) number of months customers remain subscribed before churning, calculated separately for each service category using the Kaplan-Meier estimator (which correctly accounts for censored/still-active customers rather than ignoring them).

Layman meaning: The ranking is consistent and clear: Basic Service customers churn fastest (median 22 months), followed by E-Service (45 months), then Total Service (61 months), with Plus Service customers lasting longest (median 69 months) before churning. Note that mean and median disagree for some tiers (e.g. Basic Service mean=48.4 vs median=22) — this happens because a subset of long-tenured, still-active (censored) customers pulls the average upward, while the median gives a more honest picture of the 'typical' customer's actual tenure.

Actionable for the user: Always quote median tenure, not average tenure, when discussing typical customer lifespan with stakeholders — the mean is inflated by a minority of long-surviving customers and would understate how quickly a typical Basic Service customer actually churns.

Output 3 — Overall Comparison of Survival Curves Across Categories

Test	Chi-Square	df	Sig.
Log Rank (Mantel-Cox)	50.495	3	.000
Breslow (Generalized Wilcoxon)	51.573	3	.000
Tarone-Ware	52.163	3	.000

Reproduced from actual SPSS Kaplan-Meier output: Log-Rank, Breslow, and Tarone-Ware test statistics

What this is: Three separate statistical tests, each checking the same question from a slightly different angle: is the difference in survival curves between the four service categories real, or could it just be random sampling variation?

Layman meaning: All three tests agree emphatically: the differences between service tiers are statistically significant ($p < .001$ in every test). This isn't a borderline or single-test result — Log-Rank, Breslow, and Tarone-Ware are testing different parts of the survival curve (Log-Rank weights later time points, Breslow weights earlier ones), and all three still land on the same conclusion.

Actionable for the user: Because three independent statistical tests confirm the same pattern, a business can act on the tier-based churn difference with confidence — this is not a fragile, single-test finding that might not replicate.

Business Takeaway

Service tier is a strong, statistically confirmed predictor of how long a telecom customer stays before churning. Basic Service customers churn roughly three times faster than Plus Service customers (22 vs 69 months median tenure), so retention spend should be weighted heavily toward the Basic tier, with outreach timed before the ~20-month mark where churn risk concentrates.

Note on Chart Evidence

The original analysis also produced Kaplan-Meier Survival Function and Hazard Function plots (cumulative survival and hazard curves by customer category) in SPSS. These chart images were not recoverable from the uploaded document for this write-up — insert the original Survival Function and Hazard Function chart screenshots from the SPSS output file here to complete the visual evidence.